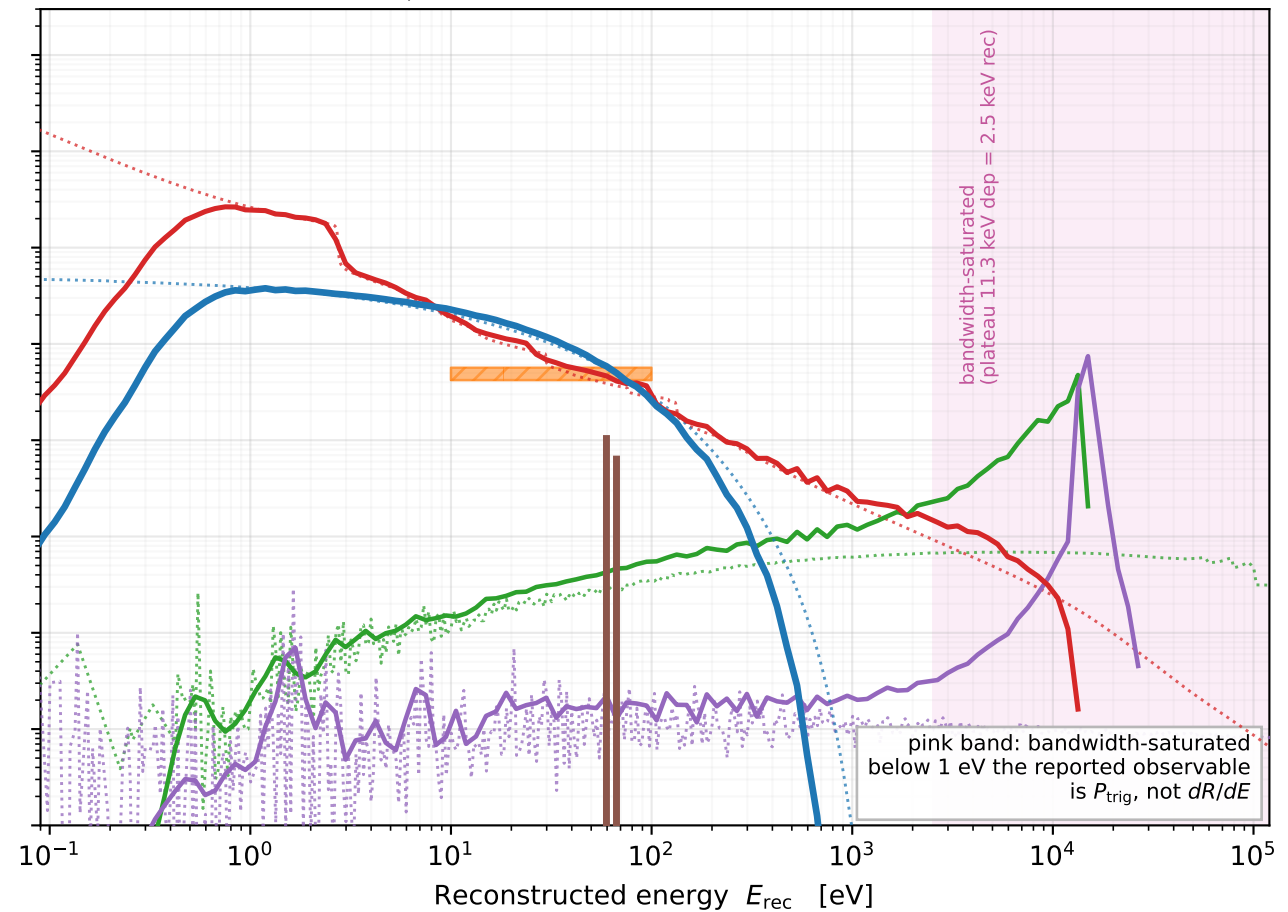
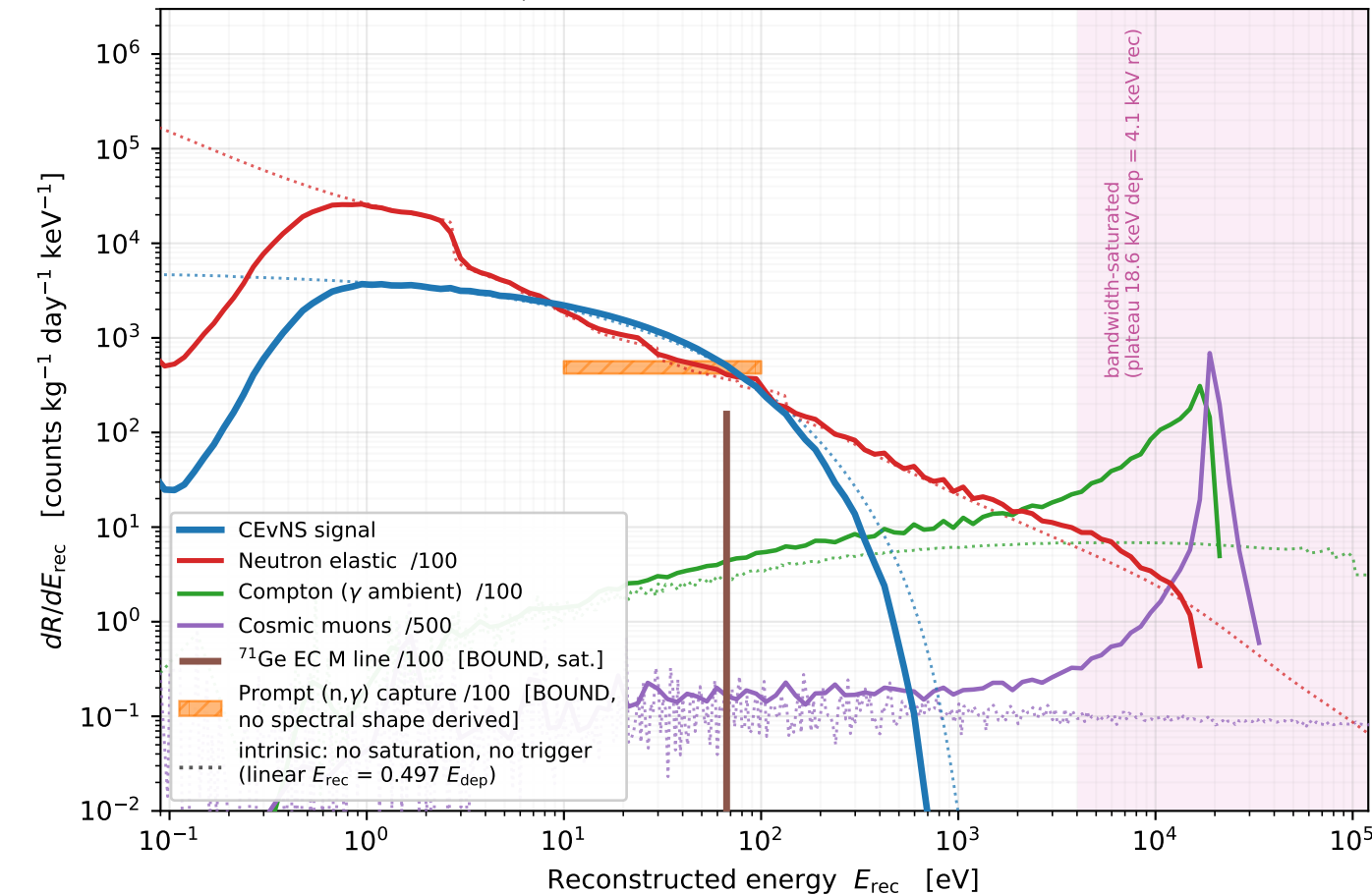


Ta → Al $S/B_{\text{particle}} = 1.334$ (est.) / 0.729 (incl. bounds)

Al → Hf $S/B_{\text{particle}} = 1.324$ (est.) / 0.728 (incl. bounds)



Per-channel factors are NUCLEUS's COMBINED shield+veto reductions (Fig. 8: >99.8% muon, ~2 orders of magnitude for the neutron-dominated residual), ASSUMED achievable by a future shield/veto — NOT rigorous credit: they are CaWO₄ event-rate reductions and most need the COV/MV veto the wafer does not geometrically fit (Phase 8).
 spectrum: linear $E_{\text{rec}} = 0.497 E_{\text{dep}}$, NO saturation and NO trigger. The solid-to-dotted gap is the trigger rolloff below ~1 eV and saturation above the onset. Dotted curves are spikier because R smooths the deposit spectra's MC noise -- that is samp